

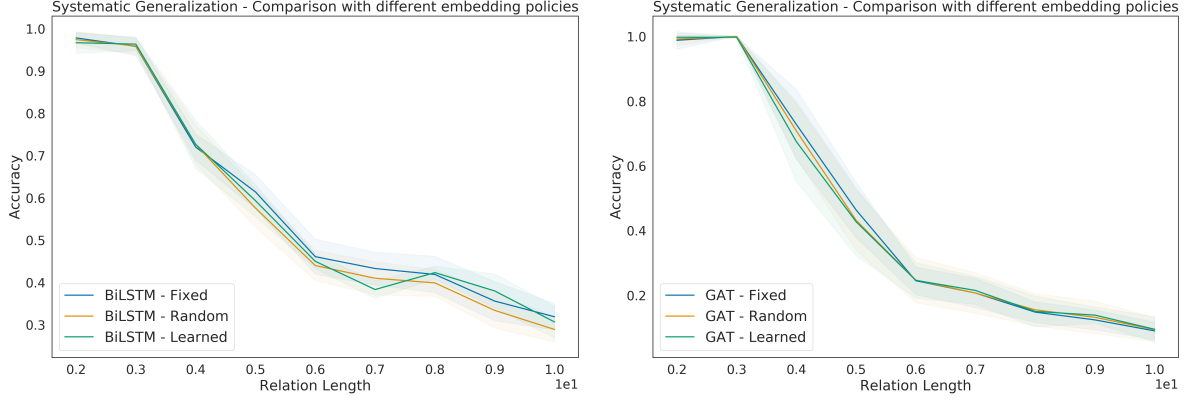


Figure 7: Systematic Generalization comparison with different Embedding policies

Relation Length	Human Performance		Reported Difficulty
	Time Limited	Unlimited Time	
2	0.848	1	1.488 +- 1.25
3	0.773	1	2.41 +- 1.33
4	0.477	1	3.81 +- 1.46
5	0.424	1	3.78 +- 0.96
6	0.406	1	4.46 +- 0.87

Table 5: Human performance accuracies on CLUTRR dataset. Humans are provided the Clean-Generalization version of the dataset, and we test on two scenarios: when a human is given limited time to solve the task, and when a human is given unlimited time to solve the task. Regardless of time, our evaluators provide a score of difficulty of individual puzzles.

To ensure that the turkers are providing high-quality annotations without revealing the inferred fact, we also launch another task to ask the turkers to rate three annotations to be either good or bad which are provided by a set of *different* turkers. We pay 0.2\$ for each HIT consisting of three reviews. This helped to remove logical and grammatical inconsistencies to a large extent. Based on the reviews, 79% of the collected paraphrases passed the peer-review sanity check where all the reviewers agree on the quality. This subset of the placeholders is used in the benchmark. A sample of programmatically generated dataset for clause length of $k = 2$ to $k = 6$ is provided in the tables 7 to 11.

1.7 Human Evaluation

We performed a human evaluation study to analyze the difficulty of our proposed benchmark suite, which is provided in Table 5. We perform the evaluation in two scenarios: first a time-limited scenario where we ask AMT Turkers to solve the puzzle in a fixed time. Turkers were provided a maximum

time of 30 mins, but they solved the puzzles in an average of 1 minute 23 seconds. Secondly, we use another set of expert evaluators who are given ample time to solve the tasks. Not surprisingly, if a human being is given ample time (experts took an average of 6 minutes per puzzle) and a pen and a paper to aid in the reasoning, they get all the relations correct. However, if an evaluator is short of time, they might miss important details on the relations and perform poorly. Thus, our tasks require *active attention*.

In both cases, we asked Turkers and our expert human evaluators to rate the difficulty of a given task in a Likert scale of 1-5, where 1 corresponds to very easy and 5 corresponds to very hard perceived difficulty. This score increases as we increase the complexity of the task by increasing the relations, thereby suggesting that a human being perceives similar difficulty while solving for larger relation tasks. However, since a human being is a systematic learner, given enough time they can solve all puzzles with perfect accuracy. We set aside the task of testing noisy scenarios of CLUTRR to human evaluators as future work.

2 Supplemental Material

To promote reproducibility, we follow the guidelines proposed by the Machine Learning Reproducibility Checklist ⁷ and release the following information regarding the experiments conducted by our benchmark suite.

2.1 Details of datasets used

A downloadable link to the datasets used can be found here ⁸. Details of the individual datasets can

⁷Machine Learning Reproducibility Checklist

⁸Dataset link in Google Drive

be found in Table 6. For all experiments, we use 10,000 training examples and a 100 testing example for each testing scenario. We split the training data 80-20 into a dev set randomly on each run.

2.2 Details of Hyperparameters used

For all models, the common hyperparameters used are: Embedding dimension: 100 (except BERT based models), Optimizer: Adam, Learning rate: 0.001, Number of epochs: 100, Number of runs: 10. Specific model-based hyperparameters are given as follows:

- **Bidirectional LSTM:** LSTM hidden dimension: 100, # layers: 2, Classifier MLP hidden dimension: 200
- **Relation Networks:** f_{θ_1} : 256, f_{θ_2} : 64, g_{θ} : 64
- **MAC:** # Iterations: 6, shareQuestion: True, Dropout - Memory, Read and Write: 0.2
- **Relational Recurrent Networks:** Memory slots: 2, Head size: 192, Number of heads: 4, Number of blocks : 1, forget bias : 1, input bias: 0, gate style: unit, key size: 64, # Attention layers: 3, Dropout: 0
- **BERT:** Layers : 12, Fixed pretrained embeddings from `bert-base-uncased` using Pytorch HuggingFace BERT repository⁹, Word dimension: 768, appended with a two-layer MLP for final prediction.
- **BERT-LSTM:** Same parameters as above, with a two-layer unidirectional LSTM encoder on top of BERT word embeddings.
- **GAT:** Node dimension: 100, Message dimension: 100, Edge dimension: 20, number of rounds: 3

⁹<https://github.com/huggingface/pytorch-pretrained-BERT>

Dataset	Variant - Training	Variant - Testing	Training Clause length	Testing Clause length
data_089907f8 data_db9b8f04	Clean - Generalization	Clean - Generalization	$(k = 2, 3)$ $(k = 2, 3, 4)$	$(k = 2, 3, \dots, 10)$
data_7c5b0e70	Clean	Clean, Supporting, Irrelevant, Disconnected	$(k = 2, 3)$	$(k = 2, 3)$
data_06b8f2a1	Supporting	Clean, Supporting, Irrelevant, Disconnected	$(k = 2, 3)$	$(k = 2, 3)$
data_523348e6	Irrelevant	Clean, Supporting, Irrelevant, Disconnected	$(k = 2, 3)$	$(k = 2, 3)$
data_d83ecc3e	Disconnected	Clean, Supporting, Irrelevant, Disconnected	$(k = 2, 3)$	$(k = 2, 3)$

Table 6: Details of publicly released data

amazon

mturk

Requester

Return

Understanding the logic behind family relations (HIT Details)

Auto-accept next HIT

Requester: Noah Turk

HITs: 29

Forecast: \$0.20

Time Elapsed: 4:07 of 30 Min

Can machines understand relations?

In this task, you will need to evaluate, simple story based on one or more facts. It is crucial that the story mentions each of the given fact(s) at least once. The story does not need to be complicated! It just needs to be grammatical and mention the required facts. For some tasks, we also provide a *hidden fact*. The story must not explicitly mention this hidden fact.

We also provide some additional information about the gender of the names. Check if the story maintains proper gender as well.

You will be asked to evaluate the quality of **three** user generated stories). It is crucial that :

- You check whether the generated story mentions each of the required fact(s)
- You check that the story does not explicitly mention the hidden fact (if it is given).
- You check the genders of the entities are correct.

Example of good and bad stories

Here we provide an example based on two facts. You may be presented with one or at most three facts in a given task.

Good Example

Facts to Mention

- John is the father of Sylvia.
- Sylvia has a brother Patrick.

Implied fact: John is the father of Patrick.

(For tasks having only one fact, there will be no implied fact.)

Written story

John is the proud father of the lovely Sylvia. Sylvia has a love-hate relationship with his brother Patrick.

Bad Example

Facts to Mention

- Vincent is the son of Tim.
- Martha is the wife of Tim.

Implied fact: Martha is Vincent's mother.

Written story

Vincent is married at Tim and his mother is Martha.

The reason the above story is bad:

- This story is bad because it is nonsense / ungrammatical.
- This story is bad because it does not mention the proper facts.

Task Manager: [3/ 3] Please rate the following story

Fact(s):

- Benjamin is William's grandson.
- Beatrice is a sister of Benjamin.

Gender information:

- Beatrice is a female.
- Benjamin is a male.
- William is a male.

The fact which is supposed to be explicitly hidden from the story:

- William has a granddaughter who is Beatrice.

Rewritten story:

William bought the latest gaming console for his grandson Benjamin this Christmas. For Benjamin's sister, Beatrice, he got a bestseller novel.

Task Manager: Please rate 1 for good and 0 for bad.

Task Manager: Thank you for your ratings. Now, please write a story using the facts below. You must mention each fact at least once.

Fact(s):

- Laura has a sister named Donna.
- Isabel has a sister named Laura.

Gender information:

- Donna is a female.
- Isabel is a female.
- Laura is a female.

And keep in mind not to reveal explicitly the following fact:

- Isabel has a sister named Donna.

Worker: Laura was excited to attend the graduation of her sister Donna. Donna was flying in from Montreal with her other sister Isabel.

Task Manager: Thank you for your help! The HIT is done now

Done with this HIT

Figure 8: Amazon Mechanical Turk Interface built using ParlAI which was used to collect data as well as peer reviews.